# Adrián Pérez Herrero

adrianperezherrero1997@outlook.es • github.com/AdrianPerezHerrero •
linkedin.com/in/adrian-perez-herrero • orcid.org/0000-0003-2085-1075
University of Santiago de Compostela, CiTIUS • A Coruña, Spain

## Education

- **PhD in Computer Science - Artificial Intelligence (Cum Laude)** — **A Coruña, Spain**
  University of Santiago de Compostela (USC) — *2021–2026*
  - Developed a modular Bayesian nonparametric framework combining hierarchical Dirichlet processes, Gaussian processes and dynamical-system modelling for clustering and characterising complex time series.
  - Applied the framework to physiological and environmental systems, including ECG analysis, respiratory-signal reconstruction and directional ocean wave spectra.
  - Developed methods for sparse environmental observations that reconstructed the source of incoming wave fronts and predicted their arrival at the Australian coastline using measurements from only two buoy sensors.
  - Contributed to the conception and implementation of a deep-learning survival-analysis framework for atrial fibrillation using irregularly sampled ECG records, combining an ODE-RNN encoder with an ODE-based decoder to model patient dynamics preceding disease onset.
- **MSc in Artificial Intelligence Research (with honours)** — **Online, Spain**
  International University Menéndez Pelayo (UIMP) and
  Spanish Association for Artificial Intelligence (AEPIA) — *2019–2020*
  - Master's Thesis: "Hidden Semi-Markov Models, an application to ECG analysis".
- **BSc in Mathematics (with honours)** — **Santander, Spain**
  University of Cantabria — *2015–2019*

## Research and Technical Expertise

Probabilistic machine learning · Gaussian processes · Bayesian nonparametrics · Dynamical systems · Time-series and spatio-temporal modelling · Deep learning · Neural ODEs / ODE-RNNs · Variational inference · Scientific Python · PyTorch · Numerical optimisation · Reproducible ML pipelines

## Publications

### Conferences

- Felicia Håkansson, Paulo Félix, Erik Bodin, Axel Gemvik, Thomas Olsson, Jesús Presedo, Per Tornvall, Carl Henrik Ek, Adrián Pérez-Herrero. **Generative machine learning applied to the electrocardiogram (ECG) to uncover relevant ECG features in distinguishing takotsubo syndrome from myocardial infarction**. *ESC (European Society of Cardiology). Online abstract supplement of the European Heart Journal - Volume 47, October 2026*.

### Preprints

- Adrián Pérez-Herrero, Paulo Félix, Jesús Presedo, Carl Henrik Ek. **Bayesian Nonparametric Dynamical Clustering of Time Series**. *arXiv preprint: 2510.06919*.
- Adrián Pérez-Herrero, Paulo Félix, Erik Bodin, Axel Gemvik, Thomas Olsson, Jesús Presedo, Per

Tornvall, Carl Henrik Ek, Felicia Håkansson. **Generative Machine Learning-Electrocardiography to Differentiate Takotsubo Syndrome from Myocardial Infarction**. *SSRN preprint: 5702401*.

- Adrián Pérez Herrero, Paulo Félix Lamas, Jesús María Rodríguez Presedo. **Parameterisation of state duration in Hidden semi-Markov Models: an application in electrocardiography**. *arXiv preprint: 2211.09478*.

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# Projects

## Research

- **SOSFood: Sustainable Optimisation for Secure Food Systems**
  Funding institution: European Commission (HORIZON EUROPE). Reference: 101134894.
  Principal supervisors: Paulo Félix Lamas.
- **XAI4SOC: Explainable Artificial Intelligence for healthy ageing.**
  Funding institution: MICINN. Reference: PID2021-123152OB-C21.
  Principal supervisors: José María Alonso Moral and Paulo Félix Lamas.
- **ADHERE-U: Models, techniques and methodologies based on artificial intelligence to improve therapeutic adherence.**
  Funding institution: MICINN. Reference: RTI2018-099646-B-I00.
  Principal supervisors: Paulo Félix Lamas and José María Alonso Moral.

## Open-Source Software

- **hdpgpc:** Open-source Python framework for hierarchical Dirichlet process–Gaussian process clustering, supporting variational inference and online/offline sequential learning. (`hdpgpc`)
- **takotsubo:** Interpretable ML pipeline for differentiating Takotsubo syndrome from myocardial infarction using ECG data. (`takotsubo`)
- **wave-spectra:** Pipeline for clustering and dynamical analysis of directional ocean wave spectra, including sparse-sensor identification of events and wave-front source/arrival forecasting. (`wave-spectra`)

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# Research Internships

- **International Research Visit** — **Cambridge, UK**
  Computer Laboratory, University of Cambridge — *2024*
  - Application of Bayesian nonparametric clustering to takotsubo discrimination.
  - Attendance and presentation at the CBCML Gaussian Process Workshop 2024.
  - Talk at Accelerate Group Meetings.
- **Undergraduate internship at CiTIUS** — **A Coruña, Spain**
  University of Santiago de Compostela — *2019*

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# Teaching

## Courses

- **Automata Theory and Formal Languages** — **A Coruña, Spain**
  Bachelor's Degree in Computer Engineering, University of Santiago de Compostela — *2023–2025*

### Bachelor's theses supervised

- Laura Salgueiro Sánchez. Nonparametric Bayesian modelling of the directional interaction on ocean wave-spectra. Bachelor's Degree in Mathematics, University of Santiago de Compostela, 2026.
- Pablo Martínez González. Breathing signal estimation from the ECG through machine learning. Bachelor's Degree in Computer Engineering, University of Santiago de Compostela, 2024.

### Master's theses supervised

- Marcos García Lamas. Characterisation of oceanographic conditions through clustering of ocean waves. Master's Degree in Big Data Analysis, University of Santiago de Compostela, 2025.
- Andrea Iturricha Artero. New metaphor visualisations for time series. Master's Degree in Big Data Analysis, University of Santiago de Compostela, 2022.

### External talks

- GPSS: Gaussian Process Summer School. Research talk: Bayesian Nonparametric Dynamic Clustering of Time Series. Manchester, 2026.

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## Professional Experience

- **Consultant – Data Analyst.** **Santander, Spain**
  LIS-Solutions. *2019–2020*
  - Business Intelligence projects, data processing (ETL), planning and stock control using Python, Java, R, KNIME, and Power BI.

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## Grants and Awards

- **Scholarship from the Xunta de Galicia**
  (grant award for PhD studies) *2022–2026*
- **Scholarship from Spanish Ministry of Education**
  (grant for BSc studies) *2015–2019*

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## Languages

Spanish (native), English (proficient user, IELTS score 8.0).

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## Additional Courses

Introduction to Blockchain coding, Solidity and Full Stack solutions. Programa de Formación e Innovación Docente (PFID) University of Santiago de Compostela. 2021-2022.
Summer School 2021 in HPC and AI. Free University of Bozen-Bolzano 2021.
Fundamentals of Deep Learning. NVIDIA. 2021.
Expert Diploma in Entrepreneurship. University of Cantabria, Santander, Spain. 2018–2019.
Volunteering work in the RSME Biennial Conference. University of Cantabria, Santander, Spain. 2019.